Hints: Probability – 2. Geometric probability and random points

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Parametrise a line by its unit normal direction $\theta \in [0, \pi)$ and signed distance $p \in \mathbb{R}$ from the origin, with measure $dp\, d\theta$ (invariant under rigid motions). For a segment of length ℓ the lines in direction θ that meet it form a p-interval of length $\ell|\cos(\theta - \theta_0)|$; integrate to get 2ℓ.

Hint 2. For a convex polygon, almost every line meeting it crosses its boundary in exactly two points. Summing over the edges, $\sum_{\text{edges}} \mu(\text{lines meeting the edge}) = 2\mu(\text{lines meeting } K)$, hence $\mu =$ perimeter; pass to general convex bodies by approximation. On the other hand $\mu = \int_0^\pi w(\theta)\, d\theta$, where $w(\theta)$ is the length of the projection on the normal direction, so the mean projection length is perimeter$/\pi$ (Cauchy).

Hint 3. Buffon: lines at spacing d, needle of length $\ell \leqslant d$. The needle crosses a line iff its projection on the normal direction covers a point of $d\mathbb{Z}$, which, given the direction, has probability (projection length)$/d$. Averaging, $\Pr = \frac{2\ell}{\pi d}$ (the mean projection of a segment is $\frac{2\ell}{\pi}$, by the previous hint with perimeter 2ℓ).

Hint 4. In $\mathbb{R}^n$ the projection of a segment of length ℓ on a random line has length $\ell|u_1|$, u uniform on S^{n-1}. Compute $\mathrm{E}|u_1|$ from the density of u_1, proportional to $(1-t^2)^{(n-3)/2}$: you get $\mathrm{E}|u_1| = \frac{\Gamma(n/2)}{\sqrt{\pi}\,\Gamma((n+1)/2)}$ (check: $2/\pi$ for $n = 2$, $1/2$ for $n = 3$).

Problem 2

Hint 1. Write both probabilities with exchangeability. With $n = 2019$ and $T = \triangle A_1A_2A_3$, $\Pr(\text{triangle}) = \binom{n}{3}\mathrm{E}\big[(|T|/\pi)^{n-3}\big]$, and $\Pr(\text{quadrilateral}) = \binom{n}{4}\Pr(Q)$, where Q is the event that $A_1, \dots, A_4$ are in convex position and all other points lie inside their hull. Note $\binom{n}{4}/\binom{n}{3} = \frac{n-3}{4} = 504$.

Hint 2. Condition on A_1, A_2, A_3. Let $R(T)$ be the set of positions of A_4 for which $A_1A_2A_3A_4$ is a convex quadrilateral (the three regions beyond a side and inside the angle at the opposite vertex). Then the quadrilateral contains T, so $\Pr(\text{quad}) \geqslant \binom{n}{4}\mathrm{E}\big[\frac{|R(T)\cap D|}{\pi}(|T|/\pi)^{n-4}\big]$. It suffices that $504\,|R(T)\cap D| \geqslant |T|$, except on negligible configurations.

Hint 3. Let $BC = a$ be the longest side, M its midpoint. Since $|OB|, |OC| \leqslant 1$, $|OM|^2 \leqslant 1 - \frac{a^2}{4}$, so the distance from M to the circle is at least $1 - |OM| \geqslant \frac{a^2}{8}$. Put Y on ray AM at distance $\frac{a^2}{8}$ beyond M: the triangle BCY lies in $R(T)\cap D$ and $|BCY|/|T| = MY/MA \geqslant \frac{a^2/8}{(\sqrt{3}/2)a} = \frac{a}{4\sqrt{3}}$.

Hint 4. So the needed pointwise inequality holds whenever $a \geqslant 4\sqrt{3}/504$; triangles with smaller longest side have area $< 10^{-4}$ and contribute at most $(10^{-4})^{2016}$ to $\mathrm{E}|T|^{2016}$, which is negligible against the contribution of triangles of area $\geqslant 1$ (positive probability), where the pointwise inequality holds with a large margin. Conclusion: the quadrilateral is more probable.

Problem 3

Hint 1. A point of the circle is never inside a polygon inscribed in the same circle, so every vertex of F is a crossing point of a red edge and a blue edge. Look at the cyclic sequence of colours.

Hint 2. Let the colours form j red runs and j blue runs around the circle. A red edge joining consecutive red points crosses the blue polygon iff a blue run lies between its endpoints, and then (if $j \geqslant 2$) exactly twice. Hence $m = 2j$ = number of colour changes if $j \geqslant 2$, while for $j = 1$ a chord separates the two hulls and $m = 0$.

Hint 3. By linearity, $\mathrm{E}(\#\text{colour changes}) = (n+k) \cdot \frac{2nk}{(n+k)(n+k-1)} = \frac{2nk}{n+k-1}$, and $\Pr(j=1) = \frac{n+k}{\binom{n+k}{k}}$. Therefore $\mathrm{E}\, m = \frac{2nk}{n+k-1} - \frac{2\,n!\,k!}{(n+k-1)!}$.

Problem 4

Hint 1. Let $\alpha_1, \ldots, \alpha_5$ be the arcs AB, BC, CD, DE, EA. Given A, the other four points are i.i.d. uniform, so $(\alpha_1, \ldots, \alpha_5)$ is uniform on the simplex $\sum \alpha_i = 2\pi$ and hence exchangeable.

Hint 2. Both triangles have base AC. Express their base angles by inscribed angles: in ACB they are $\alpha_2/2$ (at A) and $\alpha_1/2$ (at C); in ACK they are $\angle DAC = \alpha_3/2$ and $\angle ECA = \alpha_5/2$. A triangle on base AC with base angles x, y has height $AC/(\cot x + \cot y)$.

Hint 3. So the event is $\cot\frac{\alpha_1}{2} + \cot\frac{\alpha_2}{2} < \cot\frac{\alpha_3}{2} + \cot\frac{\alpha_5}{2}$. Swapping $(\alpha_1, \alpha_2) \leftrightarrow (\alpha_3, \alpha_5)$ preserves the distribution and turns the event into its complement (up to a null set). The probability is $\frac{1}{2}$.

Problem 5

Hint 1. Fix P. Since Δ lies inside Ω, the chord PQ meets Δ iff the ray PQ does, i.e. iff Q lies in the arc cut out by the two tangents from P to Δ. By the inscribed angle theorem this arc has angular size $4 \arcsin \frac{1}{|PC|}$, $C = (r, 0)$.

Hint 2. Hence the probability is $\frac{2}{\pi} \cdot \frac{1}{2\pi} \int_\Omega \arcsin \frac{1}{|P-C|}\, d\theta$, the circle average of $u(x) = \arcsin \frac{1}{|x|}$ evaluated at $x = P - C$. Study this average as a function of the centre C.

Hint 3. Check that u is subharmonic for $|x| > 1$: for $g(\rho) = \arcsin(1/\rho)$ one gets $g'' + g'/\rho = (\rho^2 - 1)^{-3/2} > 0$. The circle average of a subharmonic function, as a function of C, is subharmonic and radial, and a smooth radial subharmonic function $v(r)$ is non-decreasing (rv' is non-decreasing and vanishes at 0). The minimum is at $r = 0$.